Lab Manual
Lab 10: String Waves & Resonance

- Before the lab, read the theory in Sections 1-3 and answer questions on **Pre-lab**. Submit your **Pre-lab** at the beginning of the lab.
- During the lab, read Section 4 and follow the procedure to do the experiment. You will record data sets, perform analyses, answer questions, and have **Check Boxes** checked on **Report Sheets**. Submit your **Report Sheets** before you leave the lab.

❖ You are encouraged to have discussions with TA and other students, but you are required to do calculations and answer questions individually and independently.

1. Introduction

This lab explores oscillatory motion of a string in the form of **transverse waves**. We will generate **standing waves** on a string as a superposition of two transverse waves. We will study how the waves' physical properties, such as **wavelength**, **frequency**, and **propagating speed**, depend on the string's mass and tension. In addition, we will externally drive a spring oscillator and try to induce and observe the **resonance** phenomenon.

2. Key Concepts

- Transverse waves
- Wave speed
- Frequency and wavelength
- Standing waves and nodes
- Simple harmonic oscillation
- Resonance

3. Theory

3.1 Transverse waves on a string

As shown in Figure 1, when one shakes the end of a level string up and down, the disturbance will propagate along the string, forming **transverse waves**. The term "transverse" means that the direction of wave is perpendicular to the direction of disturbance. The shape of wave can be described by the disturbance (say y coordinate) at a given position on the string (x coordinate) and at an instant t, as $y(x,t)$. If the shaking end undergoes a simple harmonic motion, the wave becomes sinusoidal as

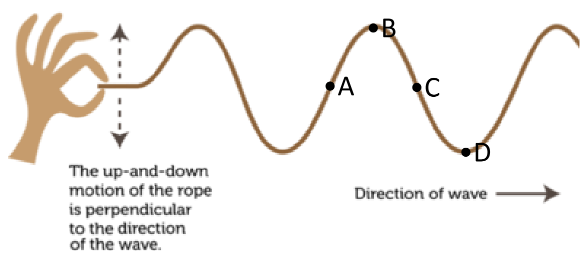


Figure 1: Generating transverse waves on a string

$$y_+(x,t) = A\sin\left(\frac{2\pi}{\lambda}x - 2\pi f t\right). \quad (1)$$

Equation (1) shows that (i) any given point of the string (at fixed x) oscillates up and down with **frequency** f, and (ii) at any given time t, the shape of the string is a sinusoidal curve with **wave-length** λ. The frequency is the number of oscillation cycles per second. The wavelength is the shortest length over which the pattern repeats. Here we use a subscript "+" to denote a wave propagating toward positive x direction. A wave toward negative x direction is written as

$$y_{-}(x,t) = A\sin\left(\frac{2\pi}{\lambda}x + 2\pi f t\right). \quad (2)$$

Regarding a point on the string, when it completes one oscillation cycle, a wave also propagates through it by one wavelength. According to the definition of frequency, the point oscillates f times per second, so the wave propagates by length of $f \times \lambda$ per second, which directly defines the wave speed as

$$v = f\lambda. \quad (3)$$

The above relation is derived from the mathematical form of sinusoidal waves. Physically, the wave speed of a string with mass m and length L depends on its tension T and linear density $\mu = m/L$ (mass per unit length), as

$$v = \sqrt{T/\mu}. \quad (4)$$

The detailed derivation can be found in the textbook. An intuitive understanding is to consider the tension as a restoring force and the mass density as inertia against it. A higher wave speed means a faster recovery of disturbance, which corresponds to a case with larger T or smaller μ.

3.2 Standing waves

As shown in Figure 2, a **standing wave** is a wave whose shape remains fixed in position over time. Standing waves can be produced by superposing two identical waves (having the same amplitude, speed, and wavelength) moving in opposite directions. In the lab, this is accomplished by fixing an end of string from which the incoming wave can almost perfectly reflect and superpose itself. The mathematical expression of a standing wave can be derived by adding together two counter-propagating waves y_+ and y_- in Eqs. (1) and (2), as

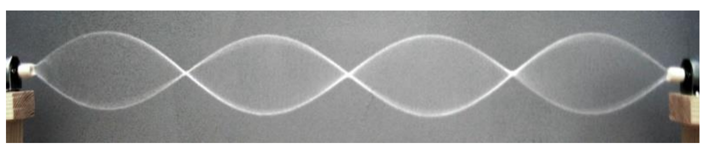


Figure 2: Standing waves on a string.

$$y_{\mathrm{SW}}(x,t) = y_+(x,t) + y_-(x,t) = 2A\cos(2\pi f t)\sin\left(\frac{2\pi}{\lambda}x\right). \quad (5)$$

We can see from the expression that while the standing wave's amplitude changes with time as $2A\cos(2\pi f t)$, its sinusoidal shape $\sin\left(\frac{2\pi}{\lambda}x\right)$ never moves.

A standing wave is often described by the number of nodes it possesses. A **node** is a place along the standing wave where the amplitude is always zero, including the endpoints in this lab. The simplest standing wave therefore has two nodes and is composed of one half of a wavelength. The second standing wave shows a full wavelength and has three nodes, and so on. We can also describe the standing wave by its **harmonic number**, which is just the number of half-wavelengths or the node number minus one.

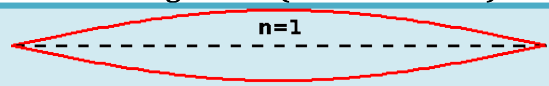
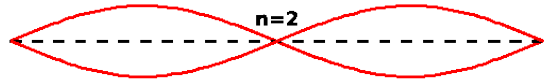
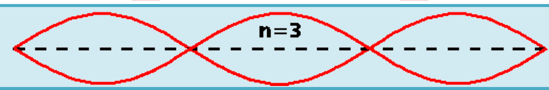
Standing waves (n= harmonic #)	**Node #**	$\boldsymbol{\lambda_n =}$	$\boldsymbol{f_n =}$
	2	$2L$	$\frac{v}{2L}$
	3	$\frac{2L}{2} = \frac{\lambda_1}{2}$	$2 \times \frac{v}{2L} = 2f_1$
	4	$\frac{2L}{3} = \frac{\lambda_1}{3}$	$3 \times \frac{v}{2L} = 3f_1$
n	$n+1$	$\frac{2L}{n} = \frac{\lambda_1}{n}$	$n \times \frac{v}{2L} = nf_1$

Table 1: Properties of standing waves on a string with fixed ends. The quantity $\boldsymbol{L}$ is the distance between both endpoints, and $\boldsymbol{v}$ is the wave speed.

Table 1 lists several properties of standing waves with different harmonic numbers. In this lab we will measure the frequencies f_n of various n, compute the wave speed as $v = 2Lf_n/n$ (why?), and compare with that predicted by Eq. (4) using the string's tension and mass density.

3.3 Resonance

When an oscillator is externally driven, its amplitude will become extremely large if the driving frequency is close to the natural frequency of the oscillator. Such peaking of amplitude is called **resonance**. In the lab, we will use the resonance phenomenon to probe the natural frequency of a spring oscillator. The setup is shown in Figure 3. We will tune frequency of the external driver (D) and observe the spring oscillator's response. At the maximum amplitude, the oscillator is on resonance, and the driving frequency f_D is equal to the spring's natural frequency,

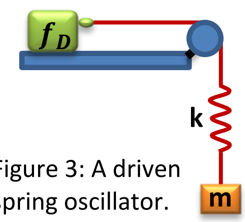


Figure 3: A driven spring oscillator.

$$f_D = \frac{1}{2\pi}\sqrt{\frac{k}{m}}. \tag{6}$$

Question (write down the answers in **Pre-lab**)

(1) Regarding the transverse wave in Figure 1, what are the moving directions of points A, B, C, and D? (answer "+" for moving up, "−" for moving down, and "0" if not moving)

(2) Apply the angle sum and difference identities on Eqs. (1) and (2) and derive Eq. (5), which is $y_+(x,t) + y_-(x,t) = 2A\cos(2\pi f t)\sin\left(\frac{2\pi}{\lambda}x\right)$.

(3) If a student varies a string's tension T and measures its wave velocity v, what regression analysis should be used to get the string's linear density μ? What do the graph's slope and y-intercept mean and how can one read out μ from them?

(4) Regarding a string with constant tension T and linear density μ, please calculate the ratio of standing wave frequency between adjacent harmonic modes f_2/f_1, f_3/f_2, f_4/f_3 and f_5/f_4.

(5) Regarding the standing wave in Figure 2, if you want to decrease the node number by 1 by changing only one of the four quantities—(i) oscillation frequency, (ii) string tension, (iii) string

linear density, and (iv) string length—how should you do it and what is your reason? (Answer “increase” or “decrease” and explain your reason.)

4. Experiment

4.1 Equipment

- Triple beam balance
- Set of masses
- Meter stick / tape measure
- Mass hanger
- Spring
- Pulley
- Short cable
- White string
- Yellow string (linear density $\mu = 1.56 \times 10^{-3}$ kg/m)
- Table clamps
- Vibration generator
- Sine wave generator
- Power adapter

4.2 Procedure

In Part I, we will find frequencies of a string’s various standing waves and verify the wave speed as a function of the string’s tension and mass density [as in Eq. (4)]. In Part II, we will induce resonance of a spring oscillator and compare the driving frequency with the oscillator’s natural frequency.

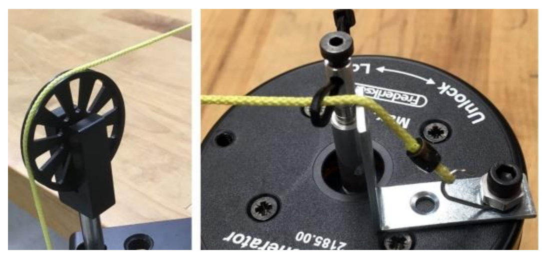


Figure 4: Standing-wave setup

Part I: Standing waves on strings

1. Measure the bare hanger’s mass and record on **Report Sheets**. **Reset the balance weights after measurement**.
2. Attach one end of the white string to the vibration generator and drape the other end over the pulley. Hang the mass hanger on the free end. The setup should look like Figure 4. Add 140 g on the hanger to stretch the string. Adjust the separation between the vibrator and the pulley such that the hanger does not touch the floor.
3. Measure the distance from the point where the string touches the vibrator to the point where it touches the pulley. Record this as the vibrating string length L in **Table R1 on Report Sheets**.
4. Switch the vibration generator to “Unlock” and connect it to the sine wave generator. Turn on the sine wave generator. Set the amplitude dial to half and the frequency to 10 Hz.
5. Replace 140 g with 60 g on the hanger.
6. Slowly adjust the frequency and amplitude of the generator until you can clearly see the first harmonic, i. e., half a wavelength. There should be nodes at both endpoints. Adjust the amplitude and fine frequency dials if necessary.
7. When you are confident that you have found the first harmonic ($n = 1$), record the frequency in **Table R1 on Report Sheets**.
8. Repeat to find frequencies for higher harmonics and for different added masses. Complete **Table R1 on Report Sheets**.
9. Repeat the experiment for the yellow string. Complete **Table R2 on Report Sheets**. (If you find that the yellow string is unstable at the pulley end, use your finger to **slightly** touch the string at this end.)

Check point 1

Ask your TA to check your data recorded and get their initials on **<u>Check Box 1</u>** on **<u>Report Sheets</u>**.

<u>Part II: Resonance of a spring oscillator</u>

1. Find the spring constant k you obtained in **Table R4** of **Lab 9 Report**. Record the value on **<u>Report Sheets</u>**. (If the spring is different from that used in your **Lab 9**, you can hang a proper mass on the spring, measure the spring extension with a ruler, and compute the spring constant by Hooke's law.)
2. Replace the string with the cable. Connect one end of spring to the cable (with a paper clip) and the other end to the mass hanger, as in the left panel of Fig 5. Put 200 g on the hanger.

Figure 5: Resonance experiment setup

3. Drape the cable over the pulley and make the vibrator horizontal such that the vibrator will drag the cable back and forth. The setup should look like Figs. 3 and 5.
4. Turn on the sine wave generator. Set the amplitude dial to half and the frequency to 3 Hz.
5. Slowly decrease the driving frequency (in units of 0.1 Hz), wait for about 15 seconds, and observe whether the spring's oscillation amplitude becomes steady or larger and larger. (Fix the vibrator by hand if it is sliding on the table.)
6. When the amplitude peaks, you have induced resonance. (If the oscillation is too drastic, turn off the driving amplitude and hold the mass in case it shoots out.)
7. Further decreasing the frequency will have the oscillator away from resonance. Fine tune the frequency back and forth until you are confident that you have found the resonant frequency. Record the frequency on **<u>Report Sheets</u>**.

Check point 2

Induce the resonance in front of TA and get their initials on **<u>Check Box 2</u>** on **<u>Report Sheets</u>**.

4.3 Analyses

I. Standing waves on strings (write on **<u>Report Sheets</u>**)

1) Compute the frequency ratio between two adjacent harmonics and compare with the theory.
2) Compute the wave speed from the standing wave frequencies and compare with the one calculated from the string tension and linear density.
3) Plot the best-fit line of $\log v$ vs $\log T$ and interpret the graph. Find the linear density from the graph and compare with the directly measured one.

Check point 3

Ask your TA to check your graphs and get their initials on **<u>Check Box 3</u>** on **<u>Report Sheets</u>**.

II. Resonance of a spring oscillator (write on **<u>Report Sheets</u>**)

1) Compare the resonant frequency with the predicted natural frequency of the spring.

- **Please clean up, recover your work station, and sign out of the computer before leaving.**